

Validation of novel *CNBP* repeat features in Myotonic Dystrophy Type 2 through comparative long-read sequencing with Oxford Nanopore and PacBio

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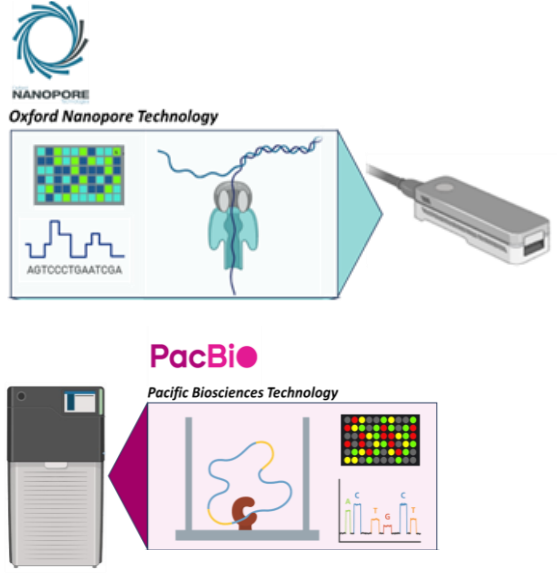


Introduction

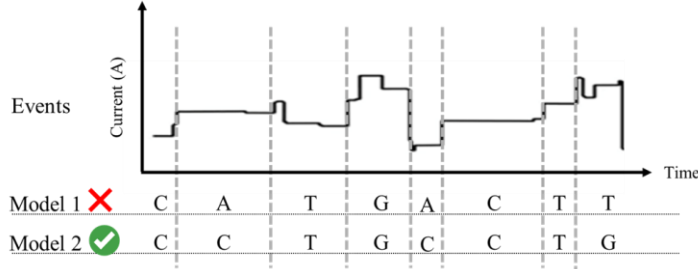
Myotonic Dystrophy Type 2 (DM2) is caused by massive and highly mosaic CCTG repeat expansions in the *CNBP* gene. While targeted long-read sequencing has enabled the sequencing of *CNBP* repeats and revealed a novel TCTG interruption pattern (Alfano M. et al, eLife 2022), the computational characterization of this region remains challenging. Currently, there is a lack of clear consensus and benchmarking of bioinformatic pipelines to accurately identify the architecture of these expansions. In this work, our goal was to establish a standardized computational framework through systematic algorithmic benchmarking and cross-platform validation, in order to accurately resolve these expansions. Additionally, we also explored the possibility to improve the basecalling accuracy by training Bonito models using PacBio data.

Methods

1. Targeted sequencing using LRS 2. Basecaller benchmarking (ONT) 3. Downstream analysis 4. Training Bonito basecaller to improve accuracy

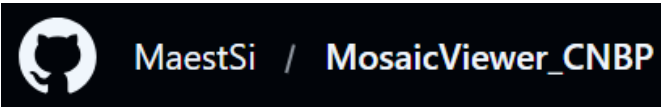


Which model and tool to use?



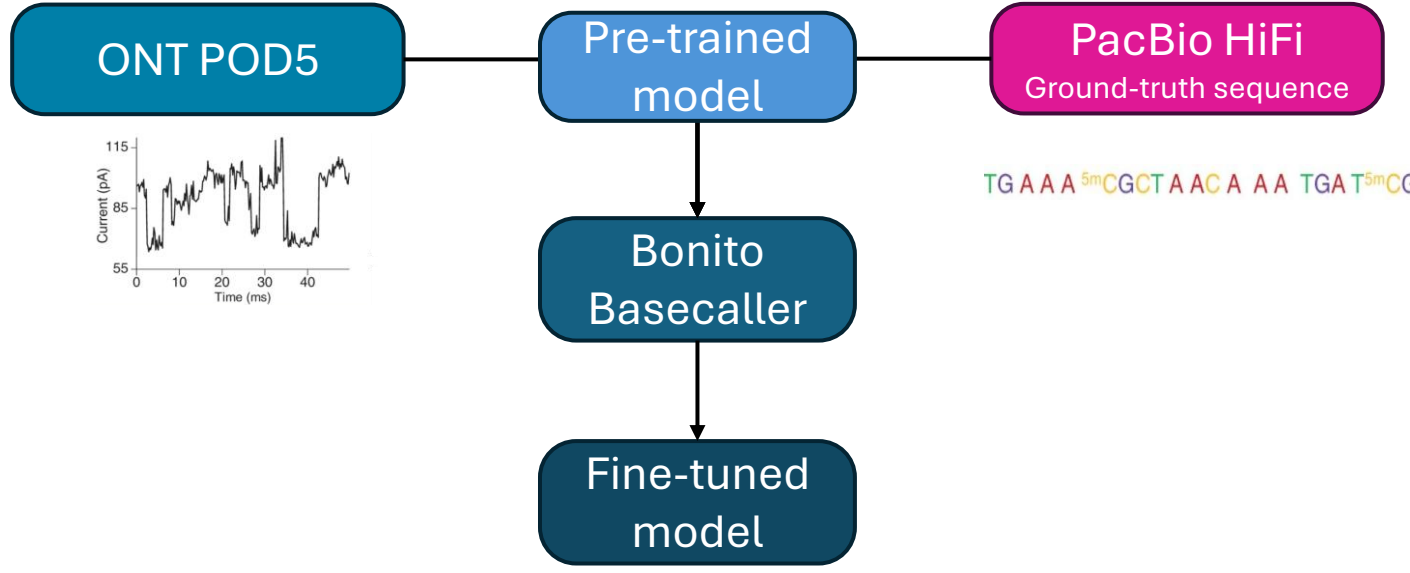
- Dorado basecaller (sup@5, sup@5.2, sup@4.3)
- Bonito basecaller (sup@4.3 untrained)
- Bonito basecaller (sup@4.3 retrained with PacBio data)

Visualization, repeat size



- Mosaicism visualization
- Maximum/median/minimum sizes alleles
- Motif quantification

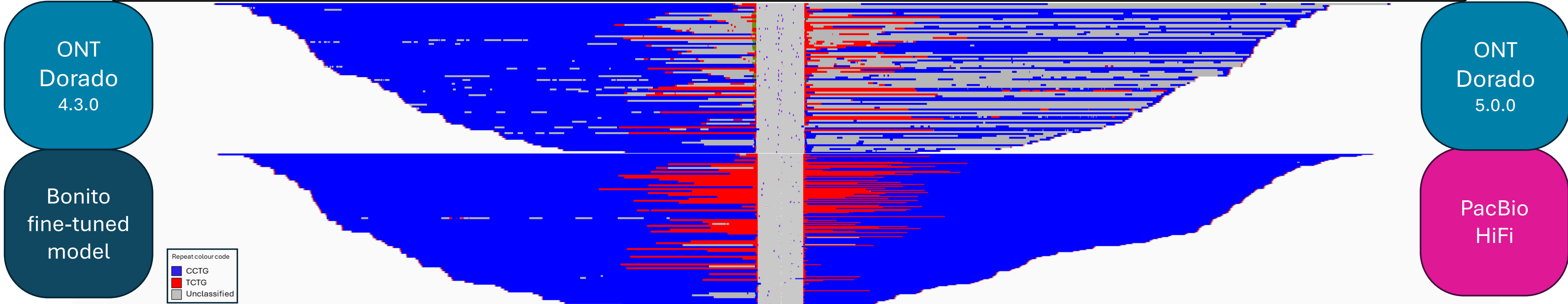
We sequenced 8 DM2 patients, using a PCR free Cas9-mediated approach combined with Oxford Nanopore (ONT) and PacBio platforms. To assess the impact of signal decoding on repeat resolution, raw POD5 files from ONT data were basecalled using multiple versions of the Dorado super-accurate model, together with the open-source Bonito basecaller. Complete reads spanning the entire expansion were extracted for downstream benchmarking. We used MosaicViewer for analyzing expansion size, motif identification, and mosaicism visualization.



In order to improve the accuracy of the models when basecalling highly repetitive motifs, we fine-tuned the Dorado model using Bonito basecaller and the HiFi sequences obtained with PacBio as ground-truth reference.

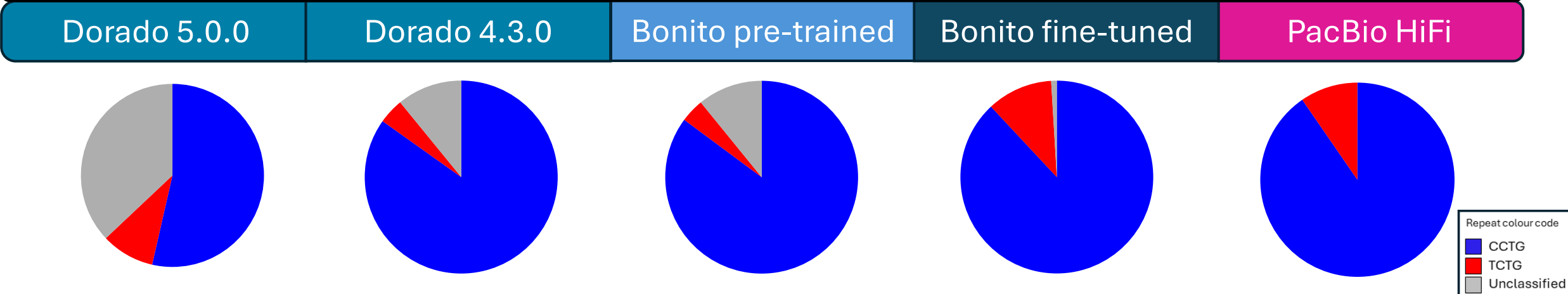
Results

Sequence structure of expanded repeats



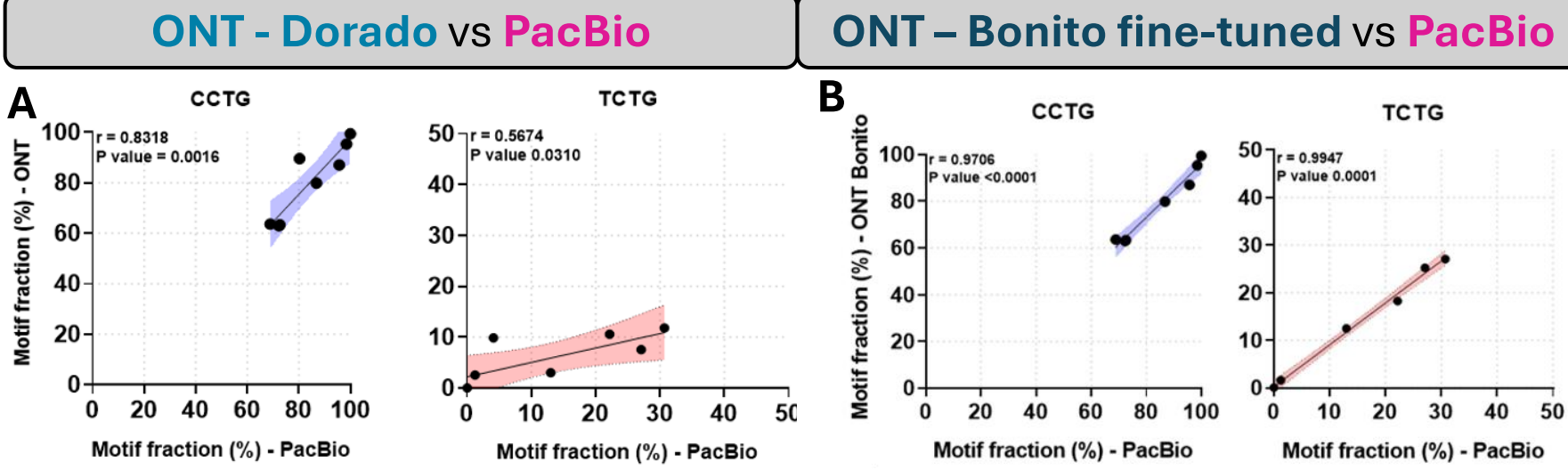
Visualization of sequencing data from the expanded alleles. Our benchmarking revealed that the choice of the basecalling algorithm version significantly impacts the accuracy of the repeat sequence identification based on ONT data. In particular, Dorado version v4.3.0 showed higher motif resolution, but still poorly resolve the TCTG motif. Fine-tuned Bonito model showed the highest concordance with PacBio High Fidelity data, used as reference control data.

Relative abundance of repeat motifs



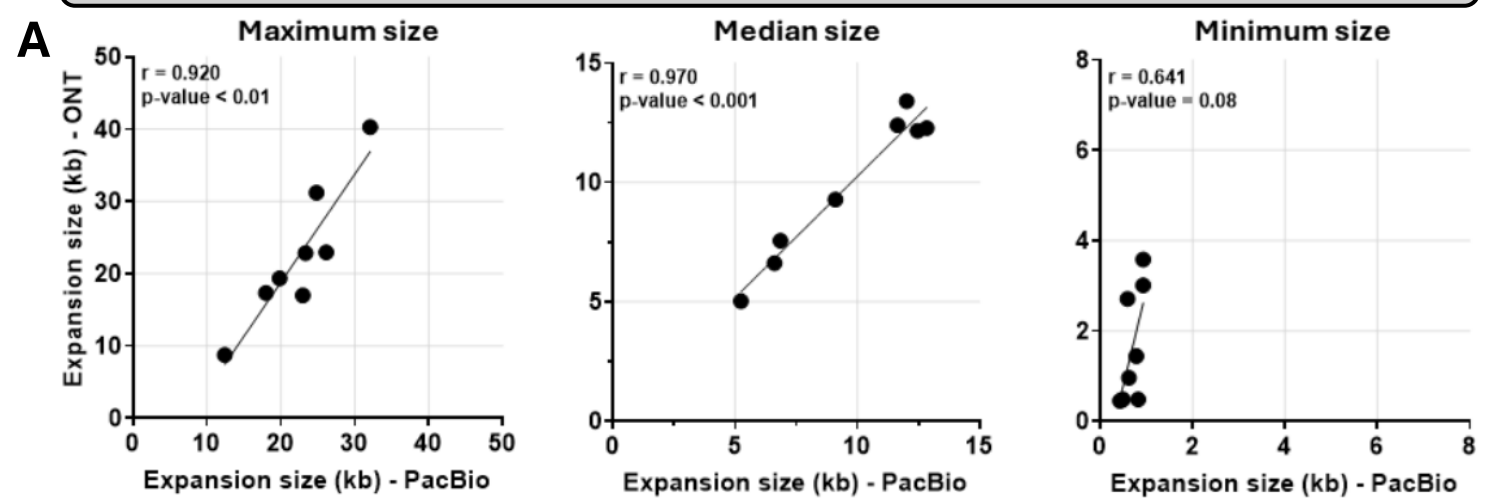
Pie charts showing the relative abundance of repeat motifs (CCTG, TCTG, CCG) within expanded reads. Selecting the appropriate Dorado basecaller model version significantly improves repeat motif resolution, and fine-tuning Bonito using PacBio HiFi data as ground-truth further reduces motif misclassification, bringing ONT results closer to PacBio accuracy.

Concordance of repeat motif in ONT vs PacBio data

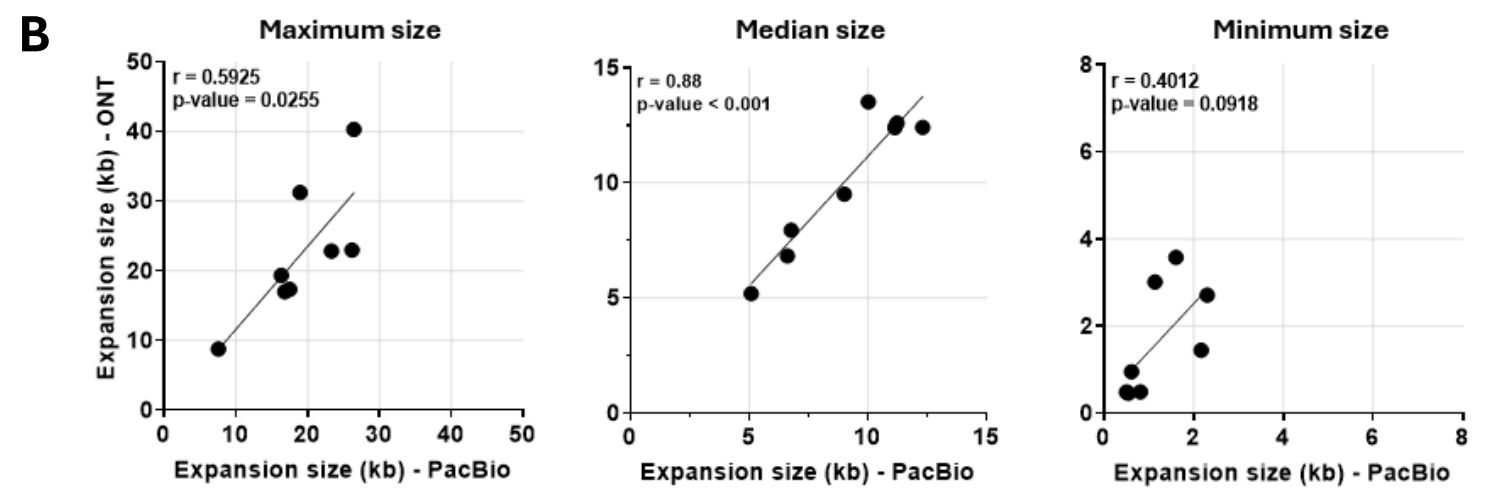


Correlations between the fraction of repeat units identified by PacBio and ONT data, either after basecalling with Dorado model v4.3 (A) or with the same model refined with Bonito (B). The latter show better concordance, obtaining similar results as PacBio.

Repeat size correlation



Repeat size correlation - after downsampling



Correlations of repeat size between ONT and PacBio data:
(A) Using the full PacBio dataset, PacBio detects shorter minimum alleles.
(B) Downsampling PacBio reads to ONT coverage improves the minimum allele size correlation, while maximum and median keeps consistent.

Conclusions

1. The basecaller model selection critically impacts the accuracy of ONT data, affecting the resolution of repeat motifs.
2. The novel TCTG motif was confirmed using PacBio data, although ONT's Dorado basecaller exhibited some limitations at identifying it.
3. Fine-tuning the ONT basecalling model improved the novel TCTG motif detection, enhancing the resolution achieved with ONT over repetitive motifs.
4. ONT and PacBio data are concordant in repeat size estimate, but sequencing coverage can influence the estimate of repeat minimum size.